

Diabetes Technology Expert Program

Trio

1. Introduction

Welcome to this comprehensive video series on Trio, an open-source automated insulin delivery system for iPhones. In these videos, we will explore the key components of Trio, including the insulin pumps and glucose sensor options, and the Oref algorithm. Additionally, we will delve into important topics such as viewing reports with Nightscout and remotely monitoring users as a caregiver.

Open-source automated insulin delivery systems are as safe and effective as commercial automated insulin delivery systems, but they have not been approved by any medical authorities. Consequently, if you choose to use it, you'll need to build and use it at your own risk. To access the manual for building the app, you can visit triodocs.org.

Additionally, you can seek assistance and support from various online communities, such as the "[Trio](#)" group on Facebook, and the "[Trio](#)" group on Discord. The Loop and Learn community also offers several explainer videos on YouTube.

Before we begin, it is essential to note that these videos are not meant as medical advice. Always consult with your healthcare provider for personalized medical advice and therapy adjustments.

The content in these videos was created and updated by dr. Inge van Boxelaer, founder of Diabetotech and endocrinologist at the St-Lucas hospital in Ghent in Belgium. The material has been reviewed by Gary Scheiner, a Certified Diabetes Care and Education Specialist from Integrated Diabetes Services in the US, and by Theresa Hastings, Trio developer and co-founder also from the US. It's important to note that Trio is available for free, and therefore, no financial interests are involved, ensuring an unbiased assessment of this open-source automated insulin delivery system.

Throughout the series, you will gain valuable insights into Trio, including its performance based on the CARES paradigm, practical usage of the Trio app, creating and interpreting reports in Nightscout, and guidance for managing various situations such as hypo and hyperglycemia, high-fat meals, exercise, illness, alcohol, and travel.

By the end of these videos, you will have a comprehensive understanding of Trio and be equipped with helpful tips and tricks to enhance your experience with the system. Whether you are a caregiver supporting Trio users or an Trio user yourself, we wish you success on your journey with this powerful automated insulin delivery system. Good luck!

2. Trio according to the CARES paradigm

In this video, we will cover Trio according to the CARES paradigm.

We will discuss how Trio calculates and adjusts the automated insulin delivery, when it reverts back to Manual Mode, specific education tips and sensor issues, and how it shares data. After that we will go over Trio indications and software updates. Let's get started!

1) Trio according to the CARES paradigm

2.1 Calculate

In Trio, the default algorithm adjusts basal insulin only. The preset basal insulin rate is adjusted based on the predicted glycemia of the next 5 to 9 hours. Here's a breakdown of how Trio works:

- Trio combines different blood glucose or "BG" predictions to calculate what the user's glucose is predicted to be at the end of the currently administered insulin's full Duration of Insulin Action or DIA :
 - o Trio calculates a carb predicted BG, which is a prediction based on the carbohydrates on board,
 - o an insulin predicted BG, which is a prediction based on the insulin on board,

- o an UnAnnounced Meal predicted BG, which is a prediction in case of unannounced carbohydrate consumption or other conditions resulting in an unexpected change in insulin needs,
 - o and a Zero Temp predicted BG, which is the worst-case scenario, in case insulin is suddenly no longer administered and remaining entered carbohydrates are no longer absorbed.
- Trio takes into account any errors in predictions, because the Eventual BG prediction is adjusted based on the deviation between the current glucose and the predicted glucose, also known as the blood glucose impact. The algorithm also considers the glycemia trend by calculating the delta, or the difference between the current and past glycemia. And it also takes into account the short-term and long-term glucose momentum, which is the amount of change between current and the glucose of 15 and 40 minutes ago.
- The insulin requirement or InsulReq is calculated by subtracting the target blood glucose from the minimal predicted blood glucose, and dividing this difference by the insulin sensitivity or ISF. This insulin requirement will determine whether to increase or decrease the manually preset basal insulin.
- If hypoglycemia is predicted, basal insulin will be stopped, and rescue carbohydrates may be recommended. This Minimum Safety Threshold is determined by the target glucose, but can be overridden to up to 120 mg/dl or 6.7 mmol/L.

In conclusion, the insulin delivery, including the baseline basal, will be raised or decreased based on the calculated insulin requirement. This is shown in the app as a temporary or temp basal with a specified insulin rate and duration or a microbolus, and these temp basals are recalculated every 5 minutes. A new calculation from Trio is also called a loop cycle.

In Trio, there is the option to deliver the insulin required calculation as a super micro bolus or the traditional temp basal. When given as a microbolus, the same insulin dose is delivered faster, followed by a temporary pause or reduction in basal insulin

delivery. How these super micro boluses or SMBs are calculated based on the temp basals, can be adjusted within predefined safety settings.

Trio also uses an autosens feature, which increases or decreases the insulin delivery, based on a comparison of the user's insulin sensitivity from the past 8 and 24 hours.

- It calculates a percentage that reflects how the insulin sensitivity in the past hours differs from the currently set sensitivity, and uses that % to adjust your insulin sensitivity and basal insulin rate.
- The Autosens function can also be applied to the target value, adjusting it based on the calculated sensitivity ratio.
- The max and min Autosens ratio are set at 120% and 70% respectively. They determine the maximum and minimum adjustment percentages based on the Autosens function. This means that the values, and thus insulin delivered, can be adjusted by no more than 120% and no less than 70% using Autosens.
- If you activate the UnAnnounced Meal function, the algorithm will detect fast glucose increments caused by carbs, adrenaline, etc., and tries to adjust it with SMBs. This also works the opposite way: if there is a fast glucose decrease, it can stop SMBs earlier.

Additionally, Trio offers Dynamic ISF which is a more aggressive way to adjust your insulin sensitivity.

- If you enable Dynamic ISF, it replaces the Autosens percentage calculation.
- It adjusts your ISF, but in a dynamic way, based on your current glucose reading. There is an inverse adjustment to ISF. The higher your glucose reading, the lower your ISF, and vice versa..
- The default formula for Dynamic ISF uses a logarithmic scale to determine the ratio for ISF adjustment.
 - This calculation uses a combination of the Total Daily Insulin from the last 2 hours and the last 10 days, current glucose reading, and accounts for the insulin type being used to determine what the current insulin sensitivity is for this loop cycle. This is considered the more physiologically accurate method for determining insulin sensitivity based on glucose readings.

- Logarithmic Dynamic ISF must first be manually tuned at the time it is enabled. The Logarithmic Adjustment Factor should start at approximately $0.02 \times$ your average total daily insulin dose. Adjustment Factor is a very individual value and will be different for each person.
- Logarithmic's aggressiveness can be adjusted using the Autosens Maximum and Minimum settings. Logarithmic Adjustment Factor should not be used to change aggressiveness.
- The second option for Dynamic ISF uses a Sigmoid formula.
 - This formula is simpler and contains fewer variables, but is less physiologically accurate. Sigmoid does not utilize your Total Daily Insulin, but relies solely on how far from target your current glucose reading is. With Sigmoid, Dynamic ISF uses the profile ISF when you are at your target glucose.
 - Sigmoid's aggressiveness can be adjusted using 3 settings: Sigmoid Adjustment Factor, Autosens Maximum, and Autosens Minimum. In Sigmoid, the Adjustment Factor should never be greater than 100% and should stay around 50% for most users. Under 50% will cause a slower response, above 50% causes a more rapid response.
 - If a user is unable to tune their Logarithmic settings, Sigmoid is the recommended option. Sigmoid is also a great option for users with rapid variability in insulin sensitivity, such as women who have variations during their menstrual cycle.

The target glucose value of Trio can be adjusted between 72 and 180 mg/dl or 4 and 10 mmol/L.

Trio places a high priority on safety and incorporates various safety settings and limits to prevent potentially dangerous insulin delivery. To ensure users configure the system correctly, the user community proposes defensive settings, and has disabled Dynamic ISF for the first 7 days of closed loop use. Over the course of several weeks, you can gradually move towards more aggressive settings.

2.2 Adjust

In Trio, you have the flexibility to adjust various parameters to personalize your treatment.

- You can set a single target glucose value per time block of 30 minutes between 72 and 180 mg/dl (4 and 10 mmol/l). International guidelines suggest starting with a target value between 110 to 120 mg/dl (or 6 to 6.5 mmol/l). Many users opt for a target value of 100 mg/dl (or 5.6 mmol/l) during the day and even lower during the night - for example 90 mg/dl (or 5 mmol/l). This can be considered if this doesn't cause significant hypoglycemia.

Carbohydrate Ratio, Insulin Sensitivity, and Insulin Peak Time can be adjusted according to your specific requirement.

- The duration of insulin action in open-source closed-loop systems is typically longer than in traditional insulin models due to the consideration of insulin's long tail effect and the impact adjusting this will have on the insulin decay model. Trio uses a Duration of Insulin Action of 10 hours as the internal formula used for insulin decay most closely matches the reported decay from insulin manufacturers.
- Trio allows you to select the type of insulin you use. You can choose from options such as Apidra, Humalog, Novorapid or NovoLog, Fiasp and Lyumjev. Depending on the selected insulin type, Trio will provide a corresponding profile. Additionally, while Duration of Insulin Action should not be adjusted, Insulin Peak Time can vary by person and this setting can be fine tuned, if necessary.

Within Trio, there are numerous other adjustable parameters, and we will delve into the primary ones: SMB options, Recommended Bolus Percentage, Autosens options, Dynamic ISF and Delivery limits.

- When starting to use Trio in "closed loop", it is recommended to use supermicroboluses only after consuming carbohydrates. If this works well for you, you can choose to activate SMB at other times as well. If any SMB options are enabled, UAM should be enabled as well. You can make the SMBs more or less aggressive by adjusting the Max SMB Basal Minutes.
- In Trio, you can change how much insulin is suggested via the bolus calculator by changing the Recommended Bolus Percentage. Trio first calculates an insulin dose based on the integrated bolus calculator, which will then be multiplied by your "Recommended Bolus Percentage" to display your suggested insulin dose. Trio will deliver the remaining insulin via SMBs

as the blood sugar starts to rise. You can view the details of the bolus calculation by tapping the “info icon” next to the recommendation.

- Trio automatically adjusts your insulin sensitivity based on an Autosens feature. You can enable Autosens to also adjust your target glucose value. You can set the maximum and minimum % your ISF can be adapted. And you can allow Autosens to adjust your ISF when you set a Temporary Target.
- Dynamic ISF can be enabled via Dynamic Settings after 7 days of successful closed looping. Adjusting the aggressiveness of Dynamic ISF is dependent upon which formula is used. After the proper Logarithmic Adjustment Factor is set, Logarithmic Dynamic ISF can be adjusted using Autosens Maximum and/or Minimum. Sigmoid Dynamic ISF can be adjusted using Sigmoid Adjustment Factor, Autosens Maximum, and/or Autosens Minimum.
- Additionally, there are delivery limits in Trio that can be adjusted within a safe range based on the user's age and type.
 - o You can set your maximum insulin on board, maximum bolus, maximum basal rate, maximum carbs on board and a minimum safety threshold. The maximum insulin on board or IOB is the total amount of basal and bolus insulin allowed to be active at once. It acts as a safety cap on how much insulin Trio can deliver beyond your scheduled basal rates. In Trio, this value defaults to 0 for safety and must be increased manually before automated insulin dosing occurs. For adults, the maximum basal rate is recommended to be set at 3 to 5 times the highest basal rate.
 - o Maximum daily safety multiplier and current basal safety multiplier are additional limits for the maximum basal insulin rate, and it's generally best to leave them unchanged at the start. The maximum daily multiplier is set to 300% by default, allowing a maximum of 3 times the highest manually set basal rate. The current basal mode multiplier is set to 400%, allowing a maximum of 4 times the manually set basal rate. These can be safely increased to up to 500% if you have low basal rates compared to your overall insulin needs for carbs.

Remember that all settings and parameters should be adjusted in consultation with your healthcare provider, taking into account your individual needs and medical guidance.

2.3 Revert

In Trio, if there is no connection to the sensor or pump for more than 15 minutes, the system will revert to Manual Mode.

When the connection is lost, the pump will continue the last set temp basal. After a maximum of 120 minutes for zero temp basals or 30 minutes for increased temp basals, it will transition to Manual Mode.

In Manual Mode or Open Loop, the basal insulin delivery is set to the preset basal rate without a suspend-before-low function. Currently, Open Loop is not recommended for Trio users. Should you want to use your pump in manual mode, it is best to set Maximum Insulin on Board (Max IOB) to 0 and set Autosens Maximum and Minimum both to 100%. This will give you a suspend-before-low function without any autosens or dynamic functions.

Once the connection to the pump or sensor is reestablished, the system will automatically resume the Auto Mode, which is called closed-loop in Trio. It will take 2-3 consecutive readings after a disconnected sensor for Trio to automatically resume closed loop.

2.4 Educate

Proper education is crucial when starting an automated insulin delivery system. For general education points, please refer to the generic module.

The general education points covered were:

- Using fewer carbohydrates to correct hypoglycemia.
- Considering infusion set problems if blood glucose levels remain high without an apparent reason.
- Trusting the automated insulin delivery system and avoiding unnecessary manual interventions.
- Timing meal boluses 15 minutes before meals.
- Utilizing an Exercise Mode 1 to 2 hours before activity.
- Stopping insulin delivery if the pump is disconnected for more than 15 minutes.
- Responding to alarms and being mindful of automatic updates on mobile phones. Automatic iOS updates should be disabled because new iOS

releases may not immediately support Trio; compatibility should always be checked first.

- Seeking peer support for guidance and sharing experiences.

Trio is a powerful tool for managing diabetes, but it's important to remember that it is not a fully automated system that you can “set and forget”. It requires active monitoring and understanding of its actions. Here are some additional education points specific to using Trio:

- Active Monitoring: Trio controls your insulin delivery, so it's crucial to watch it closely and understand how it works. Be attentive to any potential errors or unexpected behavior and learn how to interpret the system's actions.
- Phone Usage: Once paired, the phone can instruct the pump to perform various actions. Your phone with Trio is to be regarded as a medical device and be handled accordingly. Avoid installing unnecessary apps or games that could introduce malware and potentially interfere with the system's operation.
- Security Measures: Install all security updates provided by Apple to keep your device protected. This helps safeguard against potential security vulnerabilities. Do not activate “Automatic Updates” or “Beta Updates” in the Software Update settings of your iPhone. Check first the support of a specific iOS release in the Trio community prior to upgrading your iPhone operating system.
- Proximity to Pump and CGM: Keep your mobile phone in close proximity to your insulin pump and CGM. It should be within the recommended distance (which is usually within 5 meters or 15 feet for the CGM) to maintain a reliable connection.
- Passwords and Backups: Keep a record of all passwords used and store a backup of your app in iCloud. Regularly exporting your settings to iCloud simplifies the reinstallation process and prevents resetting your personalized parameters.
- Spare Mobile Phone: It's advisable to have a spare mobile phone with the Trio app and CGM apps installed available for emergencies. This ensures that you have a backup device in case of any technical issues with your primary phone.

- Check basic settings first: It is recommended to close the loop and utilize the first 7 days without access to dynamic ISF to check all your basic settings like basal insulin rate, carb ratio and insulin sensitivity.
 - a. The results of the Trio system depend largely on the correctness of these basic settings, so they need to be fairly accurate.
 - b. Next slowly raise “maximum insulin on board”, to allow Trio to automatically dose insulin. This is set by default on 0 and can slowly be raised to the sum of your average meal bolus plus 3 times your highest basal rate.
 - c. Next you can enable multiple individual SMB options or SMBs Always.
- Advanced Settings: Utilize advanced settings, such as Max SMB Basal Minutes and Dynamic ISF only after you have a good understanding of the basics and have achieved stable blood glucose control. Note that these features may not work optimally for everyone.
- Healthcare Provider Access: Ensure that your healthcare provider has access to your reports generated by Nightscout or Tidepool. This enables them to stay informed about your diabetes management and make informed decisions during your medical consultations.
- Master Insulin Pump and Sensor Use: Make sure you are able to use your insulin pump and sensor properly as well.

By following these guidelines and maintaining a proactive approach to using Trio, you can optimize the system's effectiveness and ensure a safe and reliable automated insulin delivery experience.

2.5 Sensor

Trio can be linked to the Dexcom sensors G5, G6, G7 and One Plus, European FreeStyle Libre 2 and Libre 2 Plus sensors, and Medtronic Enlite sensors via older Medtronic pumps.

Since Trio can directly connect to the European FreeStyle Libre 2 and Libre 2 Plus sensors, the LibreLink app is only needed to start a new sensor in these cases.

For Dexcom sensors, you'll need to have the associated Dexcom App running on your iPhone to allow Trio to retrieve your glucose value.

Trio also accepts additional open-source sensor apps such as Nightscout as CGM or xDrip4iOS. These apps can be used to transmit sensor data to Trio over an internet connection, although the native connections to CGM are more stable. Some people choose to use these open-source apps for better alarms and connection to other devices.

In an automated insulin delivery system, continuous and accurate sensor values are essential for proper insulin dosing. If your sensor glycemia is jumpy or noisy, it can lead to incorrect insulin dosing, resulting in high or low blood glucose levels. In such cases, it is important to disable the Auto Mode until the issue is resolved. The problem may be related to the configuration of the glucose sensor or issues with the sensor or site. If necessary, replacing the glucose sensor may be required to address the problem.

If you experience jumpy or noisy sensor values, you might try to enable the “Smooth Glucose Value” feature via the CGM Device Settings.

If your sensor does not show a number or arrow, or your readings do not match your symptoms, use your meter to make diabetes treatment decisions. When in doubt, get your meter out.

If you experience a significant difference between your sensor glucose and finger prick value, although you have a stable and in range glucose, you might try to calibrate your sensor. You can do this via the Trio Settings, via the CGM menu, or directly in the sensor app.

2.6 Share

In Trio, sharing glucose data with healthcare providers and family members is possible through Nightscout, Tidepool, Apple Health and via Dexcom Share or LibreView. Sharing glucose and insulin data is only possible via Nightscout and Tidepool, with Nightscout being the preferred option. Nightscout is the only platform that supports complete Trio data and remote commands; Tidepool does not support remote dosing and requires manual uploads.

To use Nightscout, you'll need to set up a personal Nightscout website. Trio can send the data to the Nightscout website, allowing your healthcare provider to access and review your reports in real-time.

Caregivers can access the data from the personal Nightscout website through the Loop Follow app and Nightscout. This allows them to view and track your insulin and glucose information and stay updated on your diabetes management.

Caregivers can even give an insulin bolus, enter carbs and change the insulin profile remotely, via Loop Follow. To be able to use remote care, Nightscout has to be set up to use remote commands, and the remote care has to be enabled in the Trio app.

In the Trio settings, multiple iOS shortcuts are available, but iOS Bolus shortcuts must be activated by “Allowing bolusing with Shortcuts”. This could also help parents to assist their young children remotely.

Users can download reports from Nightscout and Tidepool or use tools like Nightscout Reporter to generate comprehensive reports that can be shared with healthcare providers for in-depth analysis and evaluation.

By leveraging these sharing capabilities, Trio users can facilitate effective communication and collaboration with their healthcare team and loved ones, fostering a supportive and informed diabetes management environment.

2.7 Indications

Trio is an open-source app that does not have official indications approved by the FDA or European authorities. It can be used with all rapid-acting and ultra-rapid insulin analogues.

The Trio app is not available for download from app stores. Users are required to build the app themselves at their own risk. Setting up the system requires determination and the willingness to follow directions and learn. If you don't have the technical know-how at the beginning, you will have by the end.

All the information you need can be found in the Trio documentation that is available online, and from others who have already done it – support is available via the Trio Facebook group or other forums like Discord.

Discord is mainly used for development and build topics, whereas Facebook is more used for end-user help. Both can be used in parallel, but do not repeat each problem or topic in both environments simultaneously.

Both Facebook and Discord are highly recommended platforms for interactive chat among Trio and Nightscout users, parents, and developers. It covers a wide range of subjects, from first-time user queries to more technically advanced discussions about the inner workings of Trio and the OpenAPS algorithm. The primary language is English.

Many people have successfully built Trio and are now using it entirely safely, but it is essential that Trio users:

- Understand how their insulin pump and glucose sensor works.
- Adjust its individual dosage algorithm with the help from his or her diabetes team.
- Maintain and monitor the system to ensure it is working properly.
- Build the system themselves so that they thoroughly understand how it works, or rely on a trustworthy source to download and use the Trio App from Testflight. Testflight Apps have a 90 day validity, as this iOS environment is supported by Apple for testing new Apps.

To utilize Trio, you will need a compatible iPhone that is running on iOS 17 or newer. The minimum requirements for your devices can be found in the Trio documentation.

For building, you'll need a yearly renewable Apple Developer license, which costs 99 US dollar per year. Once this license is activated you can start the building process in 2 different ways.

- You can follow the “Build Trio with Script” procedure, using Mac computer with Xcode [Trio build](#). For this method you will first have to put your iPhone and Apple Watch if applicable to Developer Mode. On your device, you can enable this via Settings and Privacy & Security.

- Or you can follow the “Build Trio with GitHub” procedure, which is also called the “browser build” as you can use any computer for this. After completion of this method, you can download the Trio app on your iPhone via Testflight. The main advantage of this method is that it is an easier method to build and update your Trio app.

Some people don’t build the Trio app themselves, but are invited to use it via a TestFlight invitation. Some healthcare professionals may be comfortable sharing the app. If your healthcare professional provides you with the app, you won’t need an Apple Developer license.

2.8 Software updates

Regular software updates are crucial for maintaining the optimal performance and accessing the latest features of Trio. The Trio team releases new versions periodically. These updates often include bug fixes and improvements to enhance the functionality of the system.

The version of Trio is mentioned on top of the Settings screen. The most recent “main” and “dev” versions are also mentioned, to indicate “newer” version(s) available.

Once every 3 months you’ll need to update your Trio app. To update Trio, follow these steps:

- It’s good practice to update your iPhone settings to iCloud. This way all your settings of the Trio will have a backup on iCloud. You can back up your iPhone via the settings.
- When building, keep the same signing keys as the previous installation.
- If you are using the “Build Trio with script” method, you’ll use Xcode to refresh the code to the new version, while your iPhone and Apple Watch are connected and powered.
- In the case of the “Browser build” method, upon successful execution of the build process, the new version needs to be installed via a Testflight download.

If you encounter any issues, stay calm and seek support from the Trio community and documentation. Remember that troubleshooting is a normal part of software usage, and with patience, you can overcome any challenges that arise.

In conclusion, staying informed and proactive with Trio is key to optimizing your diabetes management. By keeping up with software updates, following care guidelines, and addressing any issues promptly, you can ensure a smooth and effective experience. Prioritize safety, maintain accurate records, and utilize available resources for support. Embrace the possibilities that Trio offers to enhance your health and well-being.

3. Linking components together

In this video tutorial, we will guide you through the process of creating the necessary links within the Trio automated insulin delivery system. These links establish connections between various components to ensure seamless communication and data exchange.

Trio operates in the following manner:

- Glycemia is measured every 1 to 5 minutes using a glucose sensor, such as Dexcom G6, G7, Dexcom ONE Plus, FreeStyle Libre 2 or FreeStyle Libre 2 Plus.
- After starting your European Libre 2 or 2 Plus sensor with your LibreLink app, the sensor sends the glucose readings directly to your Trio app. With Dexcom sensors, the Dexcom App needs to stay active.
- Trio calculates the required insulin dosage and sends it to your insulin pump, such as Omnipod DASH, Medtrum Nano, Dana RS, Dana-i, Omnipod Eros or older Medtronic pumps.
- The insulin pump delivers the necessary insulin and provides pump data back to the Trio app.
- Trio shares the glycemia and insulin data with Nightscout, which serves as the readout program. Trio also shares this information to Apple Health and Tidepool, and the Dexcom and LibreLink app send glucose data to Dexcom Clarity and LibreView respectively.
- Additionally, the Trio app provides an Apple Watch version being installed during the same building & installation process. This allows you to view your

glucose values in real-time on your Apple Watch, give an insulin bolus, and even start and stop overrides and temp targets directly from your Apple Watch, adding convenience to your diabetes management.

In this video tutorial, we will guide you through the process of creating various links in the Trio automated insulin delivery system. Specifically, we will cover the following steps:

- How to link your glucose sensor to your Trio app.
- How to pair your insulin pump.
- The process of creating a Nightscout account, which serves as the data hub for Trio.
- How to establish a connection between the Trio app and Nightscout, ensuring the smooth transfer of glycemia and insulin data.
- And how to configure Trio on your Apple Watch.

By following these steps, you will be able to establish the necessary links within the Trio system, enabling effective automated insulin delivery functionality for diabetes management.

3.1 Linking Sensor to Trio App

Let's delve into the process of connecting the most used sensors such as the FreeStyle Libre 2, 2 Plus, Dexcom G6, G7 and ONE Plus.

For European FreeStyle Libre 2 or 2 Plus sensors, the initial step involves starting your sensor via the LibreLink app. After the one-hour warm-up period, you can make the connection between your sensor and the Trio app. Within Trio, navigate to Settings, select Devices, CGM, and then opt for the FreeStyle Libre. Tap the configuration option to select Libre 2 Direct. Here, you can start the pairing process and scan your sensor. This enables Trio to directly receive data from your FreeStyle Libre 2 or 2 Plus sensor. Once scanned, your sensor information will populate the screen, and your current sensor glucose will be visible on the Trio homescreen. Be sure to disable bluetooth permissions in the Libre app or Trio's connection will be unstable. After 14 days, the FreeStyle Libre 2 or 2 Plus sensor will stop functioning. You can start a new sensor via the LibreLink and Trio app.

If you're using FreeStyle Libre 1 sensor with an after-market transmitter you can select Bluetooth transmitter and follow the prompts in xDrip4iOS. This sensor requires a constant internet connection and is not generally recommended.

For Dexcom G6 sensors, start the sensor using the Dexcom G6 app. In the Trio app, select Dexcom G6 in the CGM menu and enter the transmitter serial number. After a two-hour warm-up, you'll receive your results. You must keep the G6 app running in the background. Sensor changes are done in the G6 app.

For the Dexcom G7 and Dexcom ONE Plus sensor, start the sensor using the Dexcom G7 or Dexcom ONE Plus app, preferably after the warm-up period of at least 30 min, to allow overlap of sensors. Within Trio, select Dexcom G7/ONE+. When readings start in the Dexcom app, Trio will automatically read the G7/ONE+ readings.

You can opt to receive sensor alerts from either the LibreLink or Dexcom app or via the Trio app through Notifications in the Settings. Trio has 3 glucose alert options. Disabled will not provide any alerts, Always will alert you to every glucose reading, and Only Alarm Limits will alert you when glucose is above or below the limits you set.. If you require more alert options, consider using your native sensor app or an open-source application like xDrip4iOS.

3.2 Linking Insulin Pump to Trio App

Most people will use Trio with an Omnipod DASH, Dana or TouchCare Nano insulin pump. If you want to use Omnipod Eros or an older Medtronic pump, you'll need to use a Riley link.

We'll show the process of linking an Omnipod DASH Pod.

- First go to the Settings, Devices, Insulin pump and choose Omnipod DASH.
- You will first be asked to set your expiration reminder, low reservoir reminder, and your insulin type.
- Then you can tap Omnipod DASH to go to the pump screen and select "Pair Pod".

The app will take you through the different steps, similar as the Omnipod DASH PDM does. First fill your Pod with insulin, then link your Pod, remove the needle guard and place the Pod to your body. Finally continue to insert the cannula.

Once your Pod is paired, you'll see a dedicated DASH pop-up in the app interface. If you tap it, you go to the pump screen, where you can access information about your Omnipod DASH pump and control its functions. The buttons are intuitive and serve as replacements for the functions performed by the Omnipod's PDM. Although the PDM is not needed if you are using Trio, it is recommended to keep it as a backup device.

For users of other insulin pumps, the app interface will display different options corresponding to the specific pump type. These features ensure that you can effectively manage and control insulin delivery through your linked pump using the Trio app.

3.3 Creating a Nightscout Website

Nightscout is not a platform to log into but a website that you build or have built, which may involve some additional costs.

A Nightscout account consists of a website URL (name) and a password to access the website.

Nightscout accounts are created as individual user accounts. There is no Nightscout platform specifically for healthcare providers.

Nightscout enables real-time tracking of your sensor and insulin data, allowing you to share it with your loved ones or healthcare provider. Additionally, Nightscout is the exclusive method for generating reports from your Trio automated insulin delivery system.

Nightscout is more than just a website for Trio data; it can connect to various devices. Nightscout acts as a central hub, receiving data from sensors and distributing it to multiple devices such as mobile phones, smartwatches, computers, and other monitoring devices. It provides diverse options to access and display glucose information, including through smartwatches, desktop widgets, wall-mounted devices, and integration with voice assistants like Alexa.

There are multiple ways to create a Nightscout website.

Nightscout is open-source software. You can find all the information on nightscout.github.io.

- You can set it up yourself using step-by-step plans available [online](#).
- Or there are firms that offer Nightscout setup services, such as T1Pal, NS10BE, Nightscout Pro, Serendipity Bio, Nightscout4u, Nightscout-easy and Opensource.clinic.
- Nightscout and its associated services are not FDA approved, except for T1Pals' service. There is no approval from European authorities, so its use is at your own risk. Nightscout appears to be GDPR and HIPAA compliant but does not claim official compliance due to the absence of auditors.

It is crucial to handle sensor glycaemia data securely and protect your Nightscout URL with a password. If you want to share your Nightscout website with a loved one or healthcare providers, it is recommended to create a separate token for them.

By following these steps, you can create a Nightscout website easily and securely, allowing you to track and share your diabetes data effectively.

3.4 Linking Trio App to Nightscout

To connect your Trio data to your Nightscout website, just head to Settings, then Nightscout, and enter your Nightscout URL and password.

Once connected, your Trio data will show up on your Nightscout website. This gives you, your family, and your healthcare provider realtime information about how you're managing your diabetes.

If you turn on "Remote Control," trusted individuals can give you insulin, input carbs, or adjust your insulin settings using Loop Follow or their Nightscout login.

3.5 Trio App on Apple Watch

Using an Apple Watch will not only allow you to follow your sensor glucose and insulin on board on your wrist, but also allow you to add carbs, bolus and select/set/stop a profile from your watch.

Trio will automatically be downloaded on your connected Apple Watch. On your Apple Watch, you can set a Trio complication in a modular watch face.

If you tap the Trio complication on your watch face, you will see a Trio screen where you can see your glucose and loop data, your insulin on board and your carbs on board. It also lets you add carbs, do a manual bolus, and set an override or a temporary target. However, the companion app only serves as a remote control for the iPhone app; connection to the CGM and pump is still done through your iPhone, and you will have to bring it with you.

In this chapter, we explored the different connections available in Trio. From glucose sensors to insulin pumps and Nightscout, these links enable seamless communication for effective diabetes management. By establishing these connections, users can fully leverage the capabilities of Trio and take control of their diabetes.

4. How Trio works

Welcome to this video on the functionality of Trio.

Controlling your insulin delivery with the Trio automated insulin delivery system is done through the Trio app which is installed on your iPhone.

The app offers two main operating modes: Closed Loop or Auto Mode and Open Loop or Manual Mode.

- In Closed Loop, Trio automatically adjusts insulin delivery every 5 minutes based on real-time sensor glucose readings.
- If Closed Loop is disabled, you are in Open Loop or Manual Mode, and only your scheduled basal will be administered without manual intervention.

In this video, we will delve into the functionality of the Trio app in general, and explore how you can use the Trio app in both Closed Loop and Open Loop Mode.

4.1 Trio app

When you use the Trio app for the first time, an onboarding wizard will help you learn about the app.

When you open the Trio app, you can see the Trio homescreen. It has four main sections: **Status Bar**, **Graph & History**, **Adjustment Status**, and **Action Buttons**.

Let's start with the Status Bar.

Here, you'll see three elements: Pump Info, Current Glucose, Current Metrics, and the Loop Indicator.

- Pump Info shows your reservoir level and how much Pod life is left.
- Your current glucose value is displayed together with the trend information.

The Loop Indicator should normally be green. If the last loop cycle was more than five minutes ago, it turns yellow. After more than ten minutes, it turns red. And when Closed Loop is disabled, the indicator appears grey.

Right below that, you'll find the Eventual Glucose — this is a glucose forecast for up to 6 hours in the future. It is typically around 4-hours.

The second row of the Status Bar shows your Insulin on Board, Carbs on Board, and your current basal rate. Keep in mind that Insulin on Board can be negative when the algorithm has reduced the scheduled basal rate.

Next, let's look at the Graph and History section.

This area displays basal adjustments, current and past activity, active insulin, active carbs, and various stats. You can customize the graph scale and additional graph options in the Settings menu.

- If you tap on "Stats," you'll see detailed statistics for glucose, insulin, looping, and meals.
- If you tap on "Info," you'll see a full chart legend.

Below that is the Adjustment Status, which tells you whether a bolus, override or temp target is currently active. This is also where you can cancel an active bolus, override, or temp target.

Finally, at the bottom of the screen, you'll see five Action Buttons.

- "Main" brings you back to the homescreen.
- "History" shows your recent treatments, meals, glucose readings, and adjustments.
- The plus icon lets you enter carbs and boluses.
- "Adjustments" is where you can start or stop an override or temp target, and manage your presets.
- And in "Settings," you can check the software version, enable or disable Closed Loop, access the Trio configuration, and find support and community resources.

Now, let's walk through the Trio Configuration menu in detail. This section includes six parts: **Devices**, **Therapy**, **Algorithm**, **Features**, **Notifications**, and **Services**.

- First, **Devices**. Here you select your insulin pump, CGM, and smartwatch, and you can access all device-specific settings.
- Next is **Therapy Settings**. In this section, you set your unit type — mg/dL or mmol/L — and define your delivery limits. You can also enter your glucose target, basal rates, carb ratios, and insulin sensitivities. All of these can be configured in 30-minute intervals if needed.
- The third section is **Algorithm**. This is where you adjust settings related to Autosens, Super Micro Bolus, dynamic decisions, target behaviour, and other algorithm parameters. The additional algorithm settings normally don't need any changes. Only adjust them if you fully understand what they do. Every parameter includes a short description, and if you tap the information icon, you'll see a more detailed explanation.
- Next, we have **Features**. Here you can configure the bolus calculator, meal settings, shortcuts, and remote control. You can also customize the user interface and app icon. This section also lets you enable anonymised data sharing through app diagnostics.
- Then we move to **Notifications**. This is where you adjust all Trio app notifications and enable the Live Activity feature on your phone. Live Activity can appear as a dynamic island on your home screen or as a live view on your

lock screen. You can also enable calendar events, which allow your glucose values to show up on your Apple Watch and in Apple CarPlay.

- Finally, **Services**. Here you can connect your Trio app with Nightscout, Tidepool, and Apple Health.

By navigating through these sections, you can effectively manage your diabetes using the Trio app.

4.2 Closed-Loop

Now that we know how the Trio app works, we will discuss how to use Trio in Closed Loop mode.

Starting Trio Closed loop mode

Once you have completed the Onboarding Wizard, paired a pump and a CGM, you can enable Closed Loop via the main settings screen.

Use a preset Temporary Target or Override

In Trio, you can set a Temp Target or an Override via the Adjustments icon.

Overrides also allow temporary adjustments of your basal rate, carb ratio, ISF and SMB settings, according to your individual needs and preferences.

Adjusting parameters

In Trio, you have the flexibility to change your basal insulin rate, carb ratio, insulin sensitivity, and target glucose.

Giving an insulin bolus

You can deliver meal insulin using the plus icon.

The Bolus Calculator interface is divided into four sections: the **dynamic glucose forecast**, the **meal entry**, the **dosing calculation**, and the **Enact Bolus** button.

At the top, you'll see the dynamic forecast graph. As you enter carbs or adjust the insulin amount, this graph updates in real time. It shifts the forecast to reflect the expected impact of your upcoming meal and insulin dose.

Suspending insulin

If you need to temporarily stop the insulin delivery, you can go to your Pump screen by tapping the pump icons at the top left of the homescreen or via the settings. From there, you can choose the option to "Suspend Insulin Delivery" for a duration ranging from 30 minutes to 2 hours. This will temporarily suspend the insulin delivery from the pump. Here you can also resume insulin delivery.

4.3 Open Loop

In rare situations, such as extremely high and false glucose readings, it may be necessary to switch to Manual Mode in Trio. In these situations, you may lose the ability to activate Dynamic ISF when you resume closed loop, due to lack of looping history.

You can go to Manual Mode by disabling Closed Loop via the Settings.

In Manual Mode, Trio still has manual operations:

- you can adjust the profile basal insulin rate, carb ratio, insulin sensitivity, and target glucose value via the Settings
- you can set a temporary basal rate in the pump menu
- And you can deliver a bolus via the plus icon.
- You will not get glucose predictions or low glucose prevention when in open loop. Only use open loop if absolutely necessary.

Alternatively, if the concern is settings-related, it's advised to stay in closed-loop and set Maximum Insulin on Board to 0. This will maintain low glucose prevention without allowing Trio to automatically administer any additional insulin above your current scheduled basal rate. It's best to go through the documentation and have stable core settings before enabling more advanced settings, like Dynamic ISF.

While it may seem daunting to adjust the system yourself, it is important to remember that you are not alone in this journey. The Trio community is there to support you every step of the way. So, have confidence, seek help when needed, and refer to the extensive documentation as you work towards achieving optimal glycemic control with Trio.

5. Creating and interpreting reports

In this video, we will explore the process of creating and interpreting reports in Nightscout, which is used in conjunction with Trio.

Previous videos have covered the steps to create a Nightscout website and connect your Trio app to Nightscout for data transmission every 5 minutes.

Now, we will delve into the features and functionality of the Nightscout interface and reports, helping you understand and leverage the valuable information they provide for managing your diabetes.

5.1 Nightscout Interface

In the Nightscout interface, the website is typically black and can be customized based on your preferences. Here are the different sections of a Nightscout website:

1. The title bar displays the name of your site on the left and includes buttons on the right, such as the menu button with various settings options. You may also find edit and audio icons, which are rarely used. Depending on your authentication status, you will also see a plus sign, where you can log treatments.
2. System information: The system information section shows the current information of all systems connected to Nightscout. It may include gray boxes or plugins, providing additional details when hovered over. In Nightscout we call these pills.
3. User and treatment information: The user and treatment information section displays information about you and your treatments. You'll find pills or abbreviations that will be explained further in the settings section.
4. Basal insulin: The basal insulin section shows your basal insulin release.
5. Sensor glucose and treatments: The sensor glucose and treatments section displays your sensor glucose readings, carbohydrate and/or insulin treatments, as well as predictions from Trio.

6. Data history: This data history section shows the last 48 hours of data.

In Nightscout, you can access various settings through the hamburger menu at the top. Here's an overview of the available options:

- Reports: Allows you to generate reports.
- Profile and Nutrition: Via Profile and Food Editor you can change settings, although the settings are usually managed through the Trio app rather than Nightscout.
- Admin Tools enable you to create access tokens for sharing data with healthcare providers or family members.
- Account settings: Account settings provide options to adjust units, time format, language, and more. You can also customize the appearance of your Nightscout account, including the display of plugins or pills. Additionally, this section explains the abbreviations used in the interface.
- Saving settings: Changes can be saved if you enter your API password. Below that, you can see the authentication status.

By exploring these settings and making appropriate adjustments, you can personalize your Nightscout experience to suit your needs.

5.2 Types of reports

When using Trio, you can access reports through your Nightscout website via the menu in the top right.

Some of the key reports you can access include:

- Day to day Reports: These shows how the algorithm adjusts basal insulin and provides insights into micro bolus delivery.
- Distribution Report: This report displays Time in Range, Time Below Range, and estimated HbA1c. The Coefficient of Variation can be calculated by dividing the standard deviation by the mean glucose level and multiplying it by 100.
- The Percentile Chart Report offers an Ambulatory Glucose Profile-like picture to analyze glucose trends.
- The Profiles Report allows you to review and record insulin settings, such as insulin duration, carbohydrate ratio, insulin sensitivity, basal insulin rate, and target value. Algorithm settings are not saved.

- And the Loopalyzer Report provides visualizations of average basal adjustments over a 24-hour period within a specified time frame. This can help identify patterns and trends in basal insulin adjustments made by the algorithm.

Alternatively, reports from Nightscout can be visualized using the Nightscout Reporter software. This free web-based platform provides a more user-friendly interface and allows you to select the desired reports, set the time period, and generate a PDF document to forward to your healthcare provider. Interesting reports available include analysis, profiles, glucose percentile chart, and daily reports.

There are three ways to discuss these reports with your healthcare provider:

1. You can provide your healthcare provider with your Nightscout URL and a secure token that you have created. This allows them to access and view the reports directly.
2. While you can't directly generate a PDF from Nightscout, you can save the reports as PDF files. Simply open the desired report, press Command+P on Mac (or "Ctrl+P" on Windows), and select "Save as PDF" as the printer option. This will create a PDF file that you can save and email to your healthcare provider if needed.
3. Another option is to use Nightscout Reporter to create more visually appealing reports that can be easily forwarded to your healthcare provider.

By effectively utilizing Nightscout and Nightscout Reporter, you can provide your healthcare provider with comprehensive reports that highlight the performance and trends of your diabetes management with Trio.

5.3 Interpreting reports

In Trio, the main parameters that can be adjusted are the target glucose value, basal insulin rate, carbohydrate ratio, insulin sensitivity, delivery limits like maximum basal and maximum insulin on board, and the SMB settings.

This can be done via the Settings.

It is important to follow a structured approach when interpreting reports and making adjustments. Here is a detailed guide:

1. Assess Glycemic Information:

- Check on the Distribution report if treatment goals, particularly Time in Range and Time Below Range, are being met over the past 2 to 4 weeks. Following international guidelines, time in range should be above 70% and time below range should be below 4%.
- If you have access to the Analysis report of Nightscout Reporter, take note of the frequency of meal bolusing, which you would expect to be on average 3 times per day.

2. Optimize AID Settings:

- Analyze the Ambulatory Glucose Profile to identify trends of hypo- or hyperglycemia. If you spot trends, try to find out on the Daily reports, if these are related to meal boluses.
- For trends related to meal boluses: assess factors such as timing, carbohydrate counting, and consider adjusting the carbohydrate ratio.
 - If postprandial hyperglycemia is an issue, also consider using a lower Temp Target before meals.
 - You can also explore the option of faster-acting insulins like Fiasp or Lyumjev.
- For trends after manual correction boluses or after hyperglycemia: change the insulin sensitivity or duration of insulin action, to adjust the aggressiveness of the algorithm.
- For trends outside of meals or boluses: you can adjust the Target glucose value per time block or change the basal insulin rate from 1 hour before and during the corresponding period with 10-20%.
- It's also important to explore possible correlations with missed boluses, exercise, alcohol or hypoglycemia.

In Trio, it is important to ensure that the basal insulin rate is appropriately set. If you notice a consistent pattern of increasing or decreasing basal insulin during a specific period of the day that cannot be explained by SMB or manual bolusing, for example in the Loopalyzer Report, the aim is to adjust the basal insulin rate at that time to align with the algorithm's requirements.

If you are an experienced Trio user, you can control the aggressiveness of the Trio algorithm's insulin delivery in several ways.

- You can adjust the max basal and max IOB settings.
 - Lowering these limits can help prevent excessive insulin administration, especially in cases of frequent hypoglycemia.
 - On the other hand, if the max IOB setting is restricting the system's ability to correct high blood glucose levels effectively, it may need to be increased. This should be done with caution.
- You can also adjust the SMBs by changing the Max SMB Basal Minutes and Max UAM Basal Minutes
- If you use Logarithmic Dynamic ISF, you can change your Autosens Maximum or Autosens Minimum. If it's not behaving as expected, ensure your Adjustment Factor is still $0.02 \times$ your total daily insulin dose. Your total daily insulin dose may have changed and your Adjustment Factor may need to be changed as well.
- If you use Sigmoid Dynamic ISF, you can change your Sigmoid Adjustment Factor, Autosens Maximum, or Autosens Minimum.

3. Guide Behavioral Recommendations: Ensure proper usage of the automated insulin delivery system in the Daily reports:

- Sensor wear time, activation of closed loop, and adherence to pre-meal bolusing can best be checked in the Day-to-day reports.
- You can also check the correct use of temporary targets and overrides, and evaluate for overcorrection of hypo- or hyperglycemia.
- Also ask about the appropriateness of alarm settings and reminders, to avoid alarm fatigue.

4. Review Pump Settings and Emergency Plans:

- Document your pump settings, including at least your basal insulin rate, carbohydrate ratio, insulin sensitivity and target glucose value.
- Establish an emergency plan and always have insulin pens with you, in case of a pump malfunction or other emergencies.

When making changes, it is important to modify only one setting at a time. After implementing a change, closely monitor the effects over a period of 1-2 weeks. It is possible that your initial adjustment may not yield the desired results, in which case, you can explore alternative options and make further adjustments accordingly. The

key is to iterate and fine-tune your settings based on the feedback provided by your Nightscout reports, ensuring optimal performance of your Trio system.

By following these guidelines and considering the provided details, you can effectively interpret reports, optimize insulin settings, and ensure proper usage of Trio for improved diabetes management.

6. Managing special situations

In this video, we will provide specific tips and recommendations for managing common situations in diabetes management while using the Trio automated insulin delivery system.

These tips are in addition to the general guidelines covered in the general AID systems module.

We will cover strategies for handling hypo- and hyperglycemia, dealing with high-fat meals, managing diabetes during physical activities, coping with illness, alcohol and planning for travel.

By implementing these tips, you can optimize your glucose control and enhance your overall diabetes management with Trio.

6.1 Hypoglycemia

With an automated insulin delivery system like Trio, you require fewer carbohydrates to treat hypoglycemia compared to traditional diabetes management, especially when you have little insulin on board.

Trio provides the ability to see your insulin on board on the home screen. If you haven't had insulin in the last few hours, a smaller dose of carbohydrates is usually sufficient.

It is recommended not to enter the carbs you consume to treat or prevent hypoglycemia. If you consume too many carbs, you can enter the excess as a carb entry, if needed.

To prevent further episodes of hypoglycemia, you can make adjustments in the Trio system to reduce insulin delivery. This can be done by using a higher temporary target value via the Temp Target button, or via an override.

In an override, you can also choose to reduce the percentage of basal rate adjustment, inversely change your carb ratio and insulin sensitivity, and turn off SMBs, which might be a good strategy if you're experiencing hypoglycemia.

It's important to note that in some cases, the algorithm may increase basal insulin even when you are experiencing low glucose levels. This could be necessary if the predicted glucose level is higher than the target value. In the majority of cases, Trio is accurate regarding its predictions.

6.2 Hyperglycemia

In the case of mild hyperglycemia, it is generally recommended to trust Trio and allow the system to work on its own. However, you can consider giving a manual correction bolus using the bolus calculator without entering carbohydrates.

To address hyperglycemia, you also have the option to increase the percentage of the basal rate adjustment in your override or temporarily set a lower target value.

In situations of prolonged and severe hyperglycemia, it is important to consider the possibility of an infusion set problem. When in doubt, it is advisable to change your infusion set or Pod. Remember the phrase "If in doubt, change it out."

It is essential to ensure that you have enough equipment on hand to carry out all necessary substitutions and replacements. Being prepared with the required supplies is crucial for effectively managing hyperglycemia and maintaining the proper functioning of your Trio system.

6.3 High-fat meals

When consuming a high-fat meal while using Trio, there are several recommended approaches you can consider:

1. In Trio, you have the flexibility to include carbs, fat and proteins into the bolus calculator. Via the settings you can see and adjust how these fats and proteins are

taken into account in the calculation of the insulin delivery. Entering fat and protein will disable UAM functionality until those carbs are absorbed.

2. You can also utilize the Reduced Bolus Option in the bolus calculator and SMBs. It must first be enabled in the settings, but once it is, you can select this option in the bolus calculator and it will further reduce your initial bolus recommendation. If enabled, SMBs will then address the delayed rise in glucose from high-fat or protein meals over the next 6 hours following a carb entry. Ensure either SMB Always, SMB after carbs, or SMB with Carbs On Board is enabled in the SMB settings menu.

3. Another option is to administer your initial typical bolus, then allow UAMs to handle any delayed glucose rise from fat and protein. You'll need to have both SMBs and UAMs enabled in settings. You also may find you need to increase your SMB or UAM Basal Minutes to accommodate this approach.

It's important to note that individual responses to high-fat meals can vary, so it may be helpful to experiment with different approaches and monitor your glucose levels closely to find the most effective strategy for managing your blood sugar levels during such meals.

6.4 Exercise

When participating in physical activities with Trio, it is advisable to implement the following strategies:

1. Most people use a dedicated override in Trio specifically designed for exercise. This override typically should incorporate a higher target glucose, and potentially a lower insulin delivery percentage to accommodate the impact of physical activity.
2. You can also just set a higher target value during exercise to mitigate the risk of hypoglycemia.
3. In Trio, you have the option to disable SMBs when using a higher target value. This prevents increasing insulin on board during exercise which is the primary cause of hypoglycemia.

4. As discussed in the general AID module, it is also recommended to reduce your mealtime insulin if you exercise within 3 hours of your meal.
5. Take advantage of the integration capabilities of Trio by connecting your dedicated Trio app to an Apple Watch. This allows for convenient monitoring of glucose levels and trend data during exercise.

It's important to remember that these recommendations serve as a starting point, and adjustments may be necessary based on individual responses to exercise. Regular monitoring, self-observation, and collaboration with healthcare professionals will help in refining insulin dosing and achieving optimal blood sugar management during physical activity with Trio.

6.5 Illness

During illness, it is generally recommended to keep Trio in Closed Loop Mode unless advised otherwise by your healthcare provider. Some people use a dedicated override during illness for example with a higher insulin percentage and a lower glucose target.

Please refer to the guidelines about illness discussed in the general module on automated insulin delivery systems.

6.6 Alcohol

When consuming alcohol while using Trio, it is important to take certain precautions to prevent hypoglycemia. Here are some recommendations:

You can adjust your target value to a higher level than usual, or set a lower % of your insulin delivery on an override, when consuming alcohol. This can help prevent the system from administering excessive insulin in response to any increase in glucose levels.

It is worth noting that simply consuming more carbohydrates is not an effective strategy to prevent hypoglycemia, as the automated insulin delivery system will compensate for any rise in glucose levels by increasing insulin delivery.

Always monitor your glucose levels closely when consuming alcohol and be prepared to make further adjustments as needed.

6.7 Travel

When traveling with Trio, it's important to be prepared and consider some specific factors. Here are some considerations for travel:

1. Ensure that you have enough diabetes supplies, including insulin, infusion sets, sensors, test strips, and backup devices, to last throughout your trip. It's always better to be prepared with extra supplies in case of any unexpected situations.

2. When traveling by air, you may need to make some adjustments. Keep in mind the following:

- The Bluetooth function of your mobile phone, pump, and transmitter can typically remain on during the flight. If you're instructed to activate Airplane Mode, there's no need to be concerned, as iPhones will still maintain Bluetooth connectivity even when Airplane Mode is enabled.
- During the flight, the Wi-Fi, 4G and 5G connectivity of your SIM card will not be available. This should not affect the operation of the Trio app as it will continue to function perfectly via Bluetooth. However, the data will not be forwarded to Nightscout until you regain a stable internet connection.
- The Trio app usually detects time zone changes automatically and adjusts the time accordingly, although depending on pump and CGM setup, jumps in time can lead to problems. However, it's a good practice to double-check the time settings in the app after arriving at your destination to ensure accurate data recording.

Remember to always comply with airline regulations and guidelines regarding the use of electronic devices, and inform the airline staff that you are wearing a medical device such as an insulin pump.

By taking necessary precautions and planning ahead, you can enjoy your travel while ensuring the continued functionality of your Trio system.

In this chapter, we covered how to manage special situations such as hypo/hyperglycemia, high-fat meals, exercise, illness, alcohol and travel with Trio. We provided recommendations and strategies to address these situations effectively. You've got this! Remember, individual responses may vary, so

experimentation and monitoring are key. Seeking support from the Trio community and healthcare providers is also valuable.

By leveraging Trio and connecting with peers, individuals can navigate these situations and improve their diabetes management effectively.

7. Case report

You've reached the end of the Trio course. By completing every chapter, you now have a thorough understanding of Trio's interface, configuration options, algorithm behaviour, and practical use. You've learned how the system adapts, how to interpret its settings, and how to guide users through the transition phase that many people experience when moving from other AID systems.

To close the module, we want to offer you a deeper, lived-experience perspective from clinicians who use Trio themselves.

Below this lesson, you'll find the link to the YouTube episode "[Trio Open-Sourced Insulin Delivery: The Good, The Challenging, and The Differences.](#)"

In this podcast-style discussion, four highly experienced diabetes clinicians from Integrated Diabetes Services — all of them certified diabetes professionals and loopers themselves — talk openly about:

- the features they personally appreciate in Trio
- why some users prefer Trio over Loop
- how autosens, SMBs, UAM, and dynamics behave in real life
- the transition challenges they faced
- how they adjusted settings to reduce lows or handle insulin resistance
- how Trio performs in day-to-day life, exercise, meals, hormones, growth spurts, and steroid bursts
- which patient groups benefit most from the system
- and their practical strategies for basal testing, tuning the algorithm, and navigating the learning curve

This conversation brings the theory you learned in the module into real clinical context — with honest insights about what works, what doesn't, and what makes Trio different from other AID systems.

Congratulations on completing the Trio course.

You now have the knowledge to understand, support, and troubleshoot this open-sourced dynamic AID system with confidence.

Whether you use Trio yourself or guide others professionally, the insights gained here will strengthen your ability to deliver high-quality diabetes technology care.

Well done.